	Scotch Student ID #			
Circle the relevant digits	0	0	0	0
	1	1	1	1
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
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	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 2 – RATE, EQUILIBRIUM & ELECTROLYSIS

Date: Monday, June 1st

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	42	

General Instructions

- Write your student number in the space provided above on this page.
- Please complete all questions in the spaces provided.

Allowed Materials

- A VCAA Chemistry Data Book has been provided.
- Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.

At the end of the task…

- Please ensure your student number has been filled in at the top of the page and that you have written your teacher's name.

Electronic Devices

Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

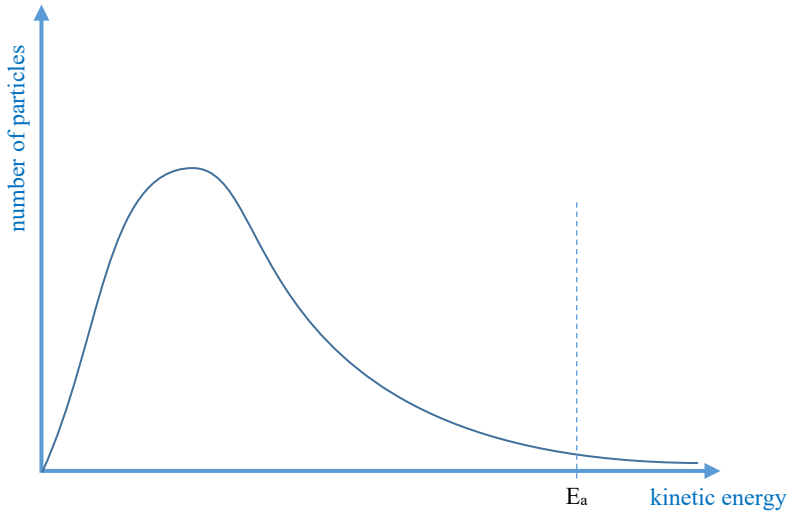
Question 1 (9 marks)

Hydrogen peroxide spontaneously decomposes into oxygen gas and water:

$$2H_2O_2(aq) \rightarrow O_2(g) + 2H_2O(l)$$

Fortunately the reaction is quite slow when refrigerated, so that hydrogen peroxide solutions can be stored safely.

The diagram at the right represents the Maxwell-Boltzmann distribution of the kinetic energies of the H_2O_2 particles in solution when refrigerated. The activation energy (E_a) for the reaction is labelled.



a. Use *collision theory* to account for the relatively low reaction rate in the refrigerator. If you wish you may label any relevant features on the diagram to assist your explanation.

2 marks

At low temperature particles are moving slowly (½)

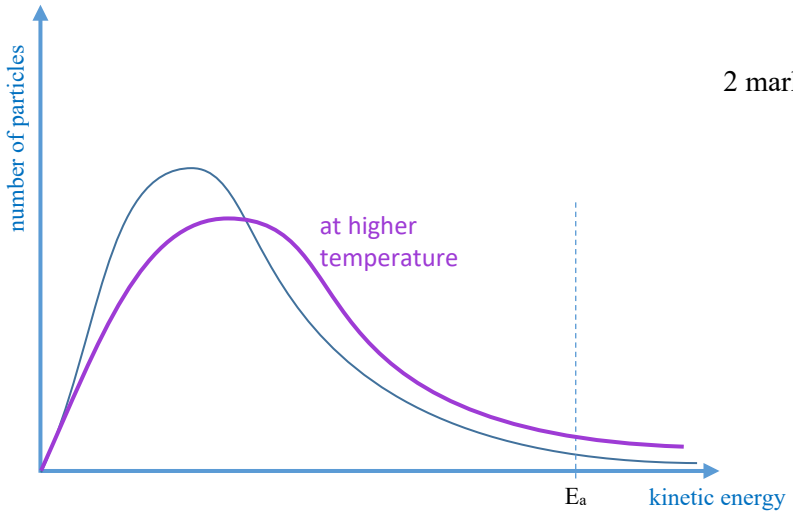
Overall ***frequency*** of collisions is low (½)

– do not pay "less" or "fewer" collisions – must be in terms of frequency or "per unit time" –as per VCAA marking scheme

Small proportion of particles have enough energy to collide and exceed activation energy requirement (1)

b. If not stored in the fridge, hydrogen peroxide decomposes much more rapidly. Redraw the part(s) of the diagram that is affected by warming the mixture, and refer to this to explain the higher reaction rate.

2 marks

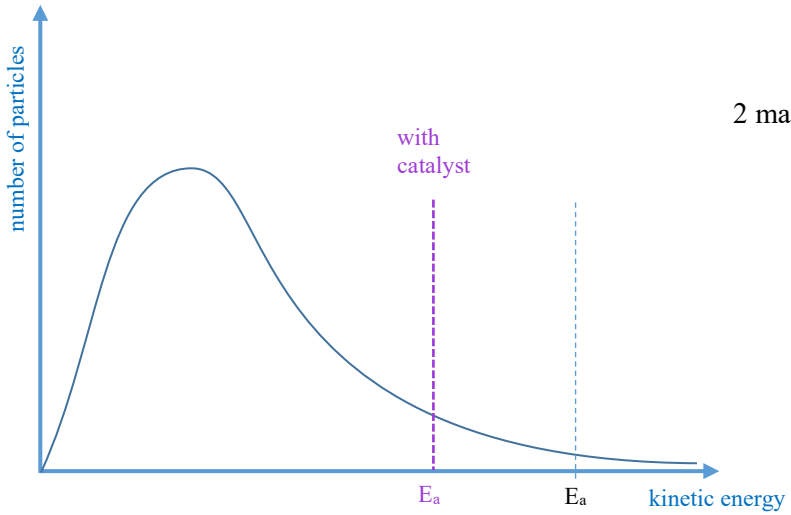


-change in shape of distribution curve (1 mk)

Particles move faster so

-higher proportion of particles with enough energy to collide and exceed activation energy requirement (1)

c. When a catalyst is added the decomposition reaction becomes fast enough to be a convenient source of oxygen gas. Redraw the part(s) of the diagram that is affected by adding a catalyst, and refer to this to explain the higher reaction rate. 2 marks



-Lower value for E_a on diagram (1 mk)

Lower E_a energy requirement, so

-higher proportion of particles with enough energy to collide and exceed activation energy requirement (1)

d. Suggest a reason that the decomposition of hydrogen peroxide in an ***open*** container would be considered an ***irreversible*** reaction. 1 mark

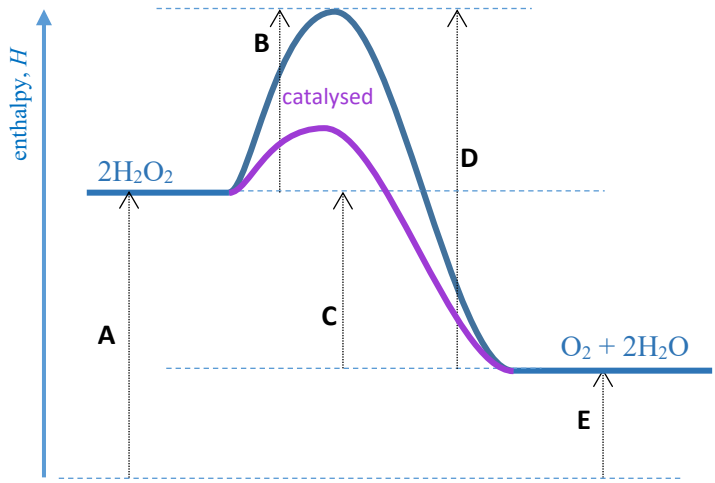
Since a gas is produced (O_2) and can escape from the reaction (½)

-the back reaction can never compete with the forward reaction/net forward reaction keeps occurring (½)

OR

Equilibrium constant is extremely high (1)

e. Below is the energy profile diagram for the decomposition of hydrogen peroxide.



i. Which of the arrows (A-E) correctly identifies the magnitude of the activation energy of decomposition reaction? 1 mark

B

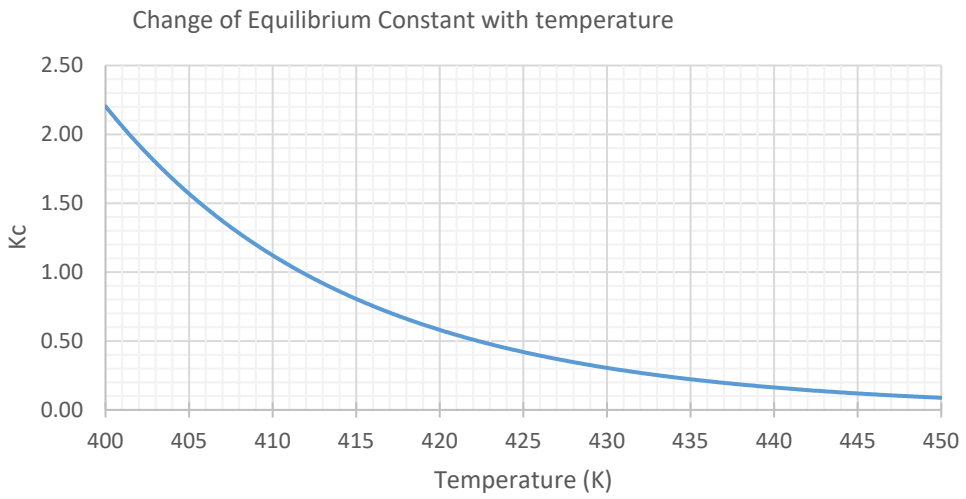
ii. On the diagram, show the effect of adding a catalyst to the reaction mixture. 1 mark

The information below relates to Question 2

Methanol is produced industrially by the gaseous reaction between carbon dioxide and hydrogen:

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

The graph below shows how the equilibrium constant for this reaction changes with temperature.



Question 2 (7 marks)

a. Write the expression for the equilibrium constant, K_c, for the reaction as represented by the equation above. 1 mark

$$K_c = \frac{[CH_3OH]}{[CO][H_2]^2}$$

b. What is value, including the correct unit, for the equilibrium constant at 415 K? 1 mark

0.80 M^{-2} (± 0.02)

c. Is the reaction represented in the equation exothermic or endothermic? Justify your answer. 2 marks

Exothermic (1) since K is decreasing as T increases (1)

Question 2 continued over the page

d. A reaction mixture, in a 750 mL steel vessel, at 415 K, was analysed and found to have concentrations as follows: $[CH_3OH] = 0.090$ M, $[CO] = 0.210$ M, $[H_2] = 0.180$ M

Show that this mixture is not at equilibrium ***and*** state the direction of the net reaction that will occur for equilibrium to be established. 2 marks

$Q = \frac{[CH_3OH]}{[CO][H_2]^2} = \frac{0.090}{0.210\ x\ 0.180^2} = 13\ M^{-2}$ (½) Since Q ≠ K the mixture is not at equilibrium (½)

Since Q > K there will be a net back reaction (1)

e. Determine the value of the equilibrium constant, at 415 K, for the following reaction, including an appropriate unit. 1 mark

$$CH_3OH(g) \rightleftharpoons CO(g) + 2H_2(g)$$

$$K_c = \frac{1}{0.80} = 1.25 = 1.3\ (½)\ M^2\ (½)$$

[conseq (½) for value that is inverse of answer to **b.** above]

Question 3 (4 marks)

During the recharging of a lead-acid cell the following reaction is occurring at one electrode:

$PbSO_4(s) + 2e^- \rightarrow Pb(s) + SO_4^{2-}(aq)$ Data: $M(PbSO_4) = 303.3$ g mol^{-1}; $M(Pb) = 207.2$ g mol^{-1}; $M(SO_4^{2-}) = 96.1$ g mol^{-1}

Note that the mass of *solid* on the electrode decreases as it is converted from $PbSO_4$ to Pb. If the charger used delivers a current of 1.80 A and is left on overnight for 14.0 hours, calculate

a. the number of moles of electrons delivered to the electrode. 2 marks

$Q\ (charge) = It = 1.80\ x\ (14.0\ x\ 60\ x\ 60) = 9.07\ x\ 10^4\ C$ (1)

$n(e^-) = \frac{Q}{F} = \frac{9.07\ x\ 10^4}{9.65\ x\ 10^4} = 0.940\ mol$ (1)

b. the decrease in the mass of the electrode. 2 marks

$n(SO_4{}^{2-})_{lost\ from\ electrode} = \frac{1}{2}\ x\ n(e^-) = 0.470\ mol$ (½)

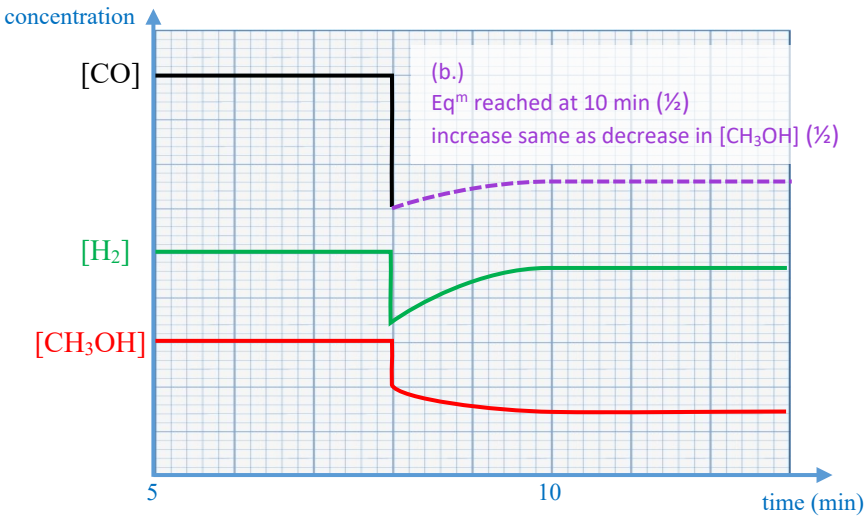
$m(SO_4{}^{2-})_{lost\ from\ electrode} = 0.470\ x\ 96.1 = 45.2\ g$ (½)

Question 4 (8 marks)

This question also relates to the reaction between carbon monoxide and hydrogen to produce methanol.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

In one reaction, the reactants were mixed and after 5 minutes had reached equilibrium. At 8 minutes a change was made, resulting in the concentration changes shown on the graph at the right. The temperature was kept constant throughout.



a. What was the likely change that was made at the 8 minute mark? 1 mark

An increase in volume (1)
[50% increase in volume]

b. Complete the graph by carefully drawing the changes expected for the concentration of CO(g) after the change shown at 8 minutes. The changes should be to scale on both axes as accurately as possible. 1 mark

c. Explain how the shift occurring in the 8-10 minute period is consistent with Le Châtelier's Principle (LCP). 2 marks

The volume increase produced a decrease in pressure (or the total gas concentration) (½)
Hence, by LCP, to oppose this change (or to increase the pressure/total concentration) (½)
there is a shift to the side with more particles (½)
ie to the left (½)

d. State how the equilibrium constant will have changed when equilibrium is reestablished after 10 min, and give a brief justification. 1 mark

No change (½) since the temperature was not changed (½)

Question 4 continued over the page

e. In a separate reaction, the graph below shows the changes in the rates of the forward and back reactions after initial mixing of the reactants. 3 marks

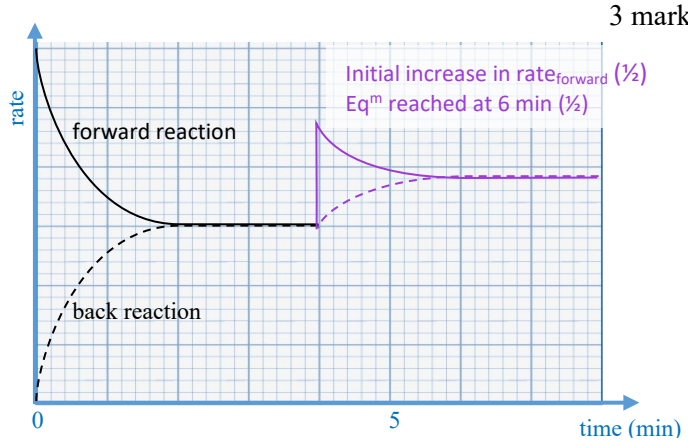
i. At what time was equilibrium reached?

2 minutes (ie when the rates become equal)

ii. After 4 min, additional hydrogen gas was added and equilibrium was reestablished at 6 min. Complete the diagram,showing the changes in the rates of the reactions between 4 and 8 minutes.

iii. How, if at all, would the concentration of methanol be different at 7 min than at 3 min?

will be higher (1)



Question 5 (4 marks)

0.715 mol of pure methanol (CH_3OH) was added to an empty 4.00 L steel vessel and heated to 100 °C, and an equilibrium was established based on the same reaction.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

At equilibrium the mixture was found to contain 0.362 mol of methanol

a. Calculate the amounts, in mol, of CO(g) and H_2(g) in the equilibrium mixture. 2 marks

	CO	$2H_2$	CH_3OH
i	0	0	0.715
c	+0.353	+0.706	-0.353
e	0.353	0.706	0.362

$n(CH_3OH)_{lost}$ = 0.715 – 0.362 = 0.353 mol (1)

$n(CO)_{formed} = \frac{1}{1}\ x\ n(CH_3OH)_{lost}$ = 0.353 mol (½)

$n(H_2)_{formed} = \frac{2}{1}\ x\ n(CH_3OH)_{lost}$ = 0.706 mol (½)

b. Calculate the equilibrium constant, K, at 100 °C, for the reaction. 2 marks

$[CH_3OH] = \frac{n}{V} = \frac{0.362}{4.00} = 0.0905\ M$

$[CO] = \frac{0.353}{4.00} = 0.0883\ M$

$[H_2] = \frac{0.706}{4.00} = 0.1765\ M$

(1)

$$Q = \frac{[CH_3OH]}{[CO][H_2]^2} = \frac{0.0905}{0.0883\ x\ 0.1765^2} = 32.9\ M^{-2}$$

(1)

Question 6 (2 marks)

Like all indicators methyl red undergoes a colour change as a result of an equilibrium shift. The equation for the equilibrium reaction is

$$C_{15}H_{14}N_3O_2^-(aq) + H^+(aq) \rightleftharpoons C_{15}H_{15}N_3O_2(aq)$$

yellow **red**

Use Le Châtelier's Principle (LCP) to predict the colour change (more yellow or more red) that occurs when potassium hydroxide (KOH) solution is added to the equilibrium mixture. 2 marks

Hydroxide will react with, and remove H^+(aq) (1)

By LCP, to partially oppose this change the equilibrium will shift to left to make more H^+(aq) (½)

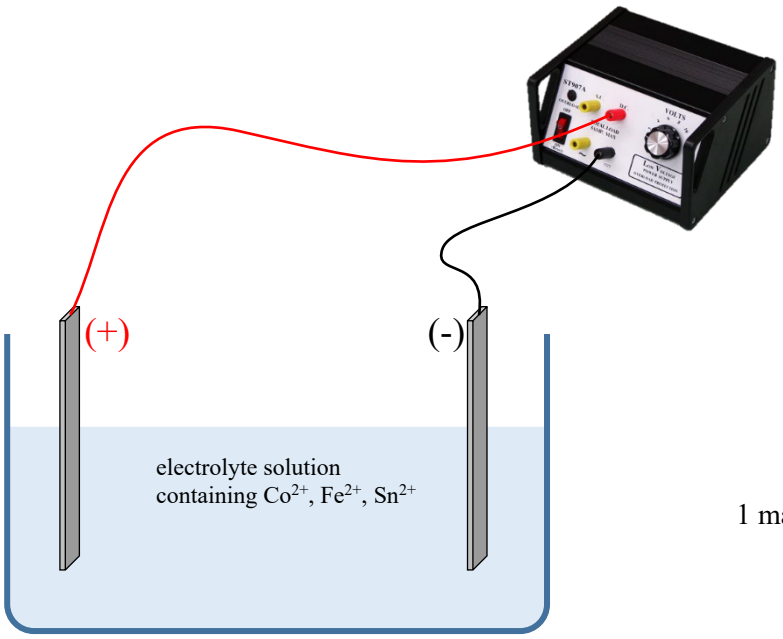
Hence the color becomes more yellow (½)

Question 7 (8 marks)

To investigate the use of electrolysis to extract cobalt from an ore deposit rich in the cobalt mineral cobaltite, a sample of the ore was dissolved in nitric acid to produce a solution of Co^{2+}(aq). Impurites in the ore included minerals of iron and tin, so the solution also contained traces of Fe^{2+}(aq) and Sn^{2+}(aq) ions.

Any excess acid in the solution was neutralised and platinum electrodes were used to pass a current through the resulting solution.

Indeed, cobalt metal was produced.



a. Identify the polarities ((+) or (-)) of each of the following in this cell. 1 mark

anode: (+) (1)

cathode: (-)

b. At which electrode ((+) or (-), or both) would cobalt metal have been produced? 1 mark

(-) (1)

c. Write the half-equation for the reaction that would produce cobalt metal in this cell. 1 mark

$Co^{2+}(aq) + 2e^- \rightarrow Co(s)$ (1)

d. Apart from the metal ions the only other significant ion present was nitrate (NO_3^-), which will not react at either electrode. Write the half-equation for the reaction that would be expected to occur at the other electrode. 1 mark

$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ (1)

e. When analysed, the cobalt metal deposited was found to be contaminated with ***one*** other metal. Use the electrochemical series to 2 marks

- determine which metal was deposited along with the cobalt
- explain why *only* this metal was deposited along with the cobalt

Only Sn will be present (1)

Sn^{2+} is a stronger oxidant than Co^{2+} (or equivalent statement) so it will be reduced (½)

Fe^{2+} is a weaker oxidant than Co^{2+} (or equivalent statement) so it will not be reduced (½)

f. Use the electrochemical series to explain why reactive metals such as sodium and aluminium are produced by electrolysis from molten mixtures or non-aqueous solutions, rather than from aqueous solutions. 2 marks

Ions of reactive metals like Na and Al (Na^+ and Al^{3+}) are weaker oxidants than H_2O. (1)

In aqueous solution water is reduced preferentially to the metal ions. (1)

END OF TASK